

Announcements

- GCP demo today
 - Ensure you have redeemed Google credits through Ed discussion post instructions
 - And downloaded Google Cloud CLI for your respective OS and installed it (<https://docs.cloud.google.com/sdk/docs/install-sdk>)
- GCP assignment out today
 - Based on deployment of a simple Java logger on the Google App Engine
 - Individual Assignment. Pretty much similar to class demo.
- REST assignment due this week
- Next class TA check-in with mentors about Project 1 progress report

CS3300 Introduction to Software Engineering

Lecture 9: Tools of the Trade #5

Google App Engine, Google APIs

Dr. Nimisha Roy ▶ nroy9@gatech.edu

Contents

Last lecture

Browser → localhost → Spring Boot → Tomcat → Controller → Service

Today

Browser → Internet → Google App Engine → Spring Boot → Controller → Service

- Why deploy to the cloud?
- Google Cloud Platform
- Compute Engine vs. App Engine
- App Engine Standard vs. Flexible
- **Demo: Deploy our Spring Boot REST API**
- Google APIs
- Maps, Routes & Places
- API authentication



Google App Engine - Intro

- A fully managed serverless platform for deploying web applications and APIs.
First Version in 2008
- You provide: Application code & Application configuration
- Google manages: Servers/infrastructure , Runtime environment, Scaling, Health monitoring, Patching/failover
- Allows you to simply deploy your code, and the platform automates everything.
 - Being "fully managed" signifies that all backend tasks like server health checks, patch management, and failovers are automatically managed by the platform.



Google App Engine vs. Google Compute Engine

Compute Engine	App Engine
Delivered as Infrastructure as a Service (IAAS)	Delivered as Platform as a Service (PAAS)

IaaS provides virtualized computing resources over the internet. It offers the infrastructure – virtual machines, storage, and networking – but leaves the rest up to the user.

- Provider manages the virtualization, servers, hard drives, storage, and networking.
- User manages applications, data, runtime, middleware, and OS.

PaaS provides a platform allowing customers to develop, run, and manage applications without dealing with the intricacies of building and maintaining the infrastructure.

- Provider manages everything from the infrastructure to the software applications.
- Users manage the application and its data.



Google App Engine vs. Google Compute Engine

Compute Engine	App Engine
Unmanaged Service by Google Cloud • our responsibility to configure, administer, and monitor the system. • Google's responsibility to ensure that resources are available, reliable, and ready to use	Managed Service by Google Cloud • your focus should be on the application only • Google will manage the resources needed to run the application.
offers the users complete control and flexibility over resources	Google manages all compute resources
Requires high expertise level, since everything needs to be installed and configured by yourself.	Very easy to use and deploy
Autoscaling is slower	Faster Autoscaling
Cheaper in the longer run and for large instances	Expensive in the longer run
Less secure than App Engine	More Secure



Google App Engine - Functions

- All infrastructure to deploy app on cloud- end to end management
- Scalability- acquires more instances automatically if the traffic of your application becomes higher
- monitoring, logging, versioning, debugging using google stack driver diagnostics
- Traffic splitting
- Integration with other GCP services - big data, storage, compute, connectivity support using google cloud
- Applications run in language specific sandboxes or in docker containers, depending on environment



App Engine Environments

- **Standard:** Run in language specific sandboxes
 - Complete isolation from OS/Disk/Other Apps
 - **V1:** Java, Python, PHP, Go (OLD versions). Restricted network access & libraries for some languages – not Java
 - **V2:** Java, Python, PHP, Node.JS, Ruby, Go (NEWER Versions)
 - More constrained - Pricing is based on instance hours.
- **Flexible-** Application instances run within Docker containers
 - Makes use of Compute engine virtual machines
 - Support ANY runtime (with built –in support for Python, Java, Node.js, Go, Ruby, PHP, or .NET)
 - Provides access to background processes and local disks
 - More flexible - Pricing is dependent on memory and virtual CPU

App Engine Demo Time!

Deploy a HelloWorld application to Cloud using App Engine Standard

- Add GCP credits to your account.
- Add Billing Information
- Create a Project on Google Cloud
- Take the Spring Boot REST application from TOT #4, which currently runs on localhost, and deploy the same application to Google App Engine
 - Create the app.yaml file to configure deployment.
 - Build your application using Maven.
 - Deploy to Google App Engine using Google Cloud CLI.
- Verify the deployment and manage billing settings.

For your assignment

- Create a SpringBoot Application
- Get 2 endpoints – name and local time
- Package project into a JAR
- Deploy using app engine

For your project

- Create a SpringBoot + <Any Frontend framework> Application
 1. Package project into a JAR and deploy using app engine OR
 2. Clone your repository in Cloud Shell and deploy from there.

Another demo notes to deploy from cloud shell and as a JAR will also be uploaded on the class website.

Help Documentation to monitor projects

- [Help documentation](#) about setting alerts to monitor projects in the Console
- You can also use the [Cost Management Resource Guide](#) to learn how to monitor and manage costs in GCP.

Google APIs

- Developed by Google which allow communication with Google Services and their integration to other services.
- Examples include Search, Gmail, Translate or Google Maps.
- Third-party apps can use these APIs to take advantage of or extend the functionality of the existing services.
- Provides functionality like analytics, machine learning (the Prediction API) or access to user data (when permission to read the data is given).
- Usage of all of the APIs requires authentication and authorization using the OAuth 2.0 protocol.



Google Maps/Routes/Places API

	Maps API 	Places API 	Routes API 
Purpose	Display and interact with geographic information	Search for places and retrieve information about them	Calculate routes, distances, and travel times
What it provides	Interactive maps, markers, map controls, geographic visualization	Place search, autocomplete, place details, categories, addresses, coordinates	Directions, routes, distance/duration, travel modes, traffic-aware routing
Project 1 connection	Display search results on the interactive map	Find locations matching the user's location, radius, and location-type filter	Optional feature for directions, trip planning, or distance/travel-time comparisons
Example	Show restaurant results as markers on a map	Find restaurants within the user's search area	Find the route from the user's location to a selected restaurant

API Authentication & Authorization

Different Google APIs support different credential mechanisms.

API Keys

- Identify the calling project/application
- Common for Maps Platform APIs
- Should be restricted by API and application/domain

OAuth 2.0

- Used when an application needs authorization to access user data
- User grants permission

Service Accounts

- Application-to-application/server credentials

Use the authentication mechanism required by the specific API.